

Mathematical Modeling of Option Pricing Using the Black–Scholes PDE and Monte Carlo Simulation

Project Report

Differential Equations and Mathematical Modeling

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Abstract

This project explores option pricing through two approaches: the analytical Black–Scholes PDE method and Monte Carlo simulation. We derive the Black–Scholes formula using techniques from our differential equations syllabus — reduction to the heat equation, method of undetermined coefficients, and Laplace transforms. We then identify the model’s key limitation — the constant volatility assumption — through the volatility smile. Finally, we introduce Monte Carlo simulation as a flexible alternative that handles real-world complexities like stochastic volatility.

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1 Introduction

1.1 What is an Option?

An **option** is a financial contract that gives the buyer the right (but not the obligation) to buy or sell an asset at a fixed price (called the **strike price** K) on or before a specific date (called **expiry** T).

Example of a Call Option:

- You buy a call option on Reliance stock with strike price $K = \text{Rs.}100$ and expiry $T = 1$ month.
- After 1 month, if Reliance is at Rs. 120, you exercise the option: buy at Rs. 100, sell at Rs. 120, profit = Rs. 20 per share.
- If Reliance is at Rs. 80, you do nothing. Your maximum loss is the premium you paid to buy the option.

1.2 The Core Question

What is the fair price (premium) of this option today? This is the problem that Black and Scholes solved in 1973, earning them the Nobel Prize.

1.3 Two Approaches in This Project

1. **Black–Scholes PDE** — Analytical solution using differential equation techniques.
2. **Monte Carlo Simulation** — Numerical approach by simulating thousands of possible future stock prices.

2 The Black–Scholes PDE: Derivation and Solution

2.1 Modeling the Stock Price

The stock price S_t is modeled as **Geometric Brownian Motion**:

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (1)$$

where:

- μ = expected return (drift)
- σ = volatility (standard deviation of returns, measures uncertainty)
- W_t = Wiener process (random noise, Brownian motion)

Syllabus Connection — Week 1 (Picard’s Theorem): Picard’s theorem guarantees that equation (1) has a unique solution for any given initial stock price S_0 . The Lipschitz condition is satisfied because $|\mu S| + |\sigma S| \leq C(1 + |S|)$.

2.2 From SDE to PDE

Let $V(S, t)$ be the option price. Using **Ito's Lemma** (the stochastic version of the chain rule):

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW_t \quad (2)$$

The Hedging Trick: Construct a portfolio that eliminates randomness:

$$\Pi = -V + \frac{\partial V}{\partial S} S \quad (3)$$

The change in this portfolio cancels the dW_t term:

$$d\Pi = - \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt \quad (4)$$

Since this portfolio is now risk-free, it must earn the risk-free rate r (no arbitrage): $d\Pi = r\Pi dt$. Substituting gives the **Black-Scholes PDE**:

$$\boxed{\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0} \quad (5)$$

Terminal Condition: For a European call option, at expiry $t = T$:

$$V(S, T) = \max(S - K, 0) \quad (6)$$

2.3 Reduction to the Heat Equation

The PDE (5) has non-constant coefficients (S and S^2). We transform it to the heat equation.

Step 1: Change of Variables.

$$S = Ke^x, \quad t = T - \frac{2\tau}{\sigma^2}, \quad V(S, t) = Kv(x, \tau) \quad (7)$$

This converts the PDE to a constant-coefficient form:

$$\frac{\partial v}{\partial \tau} = \frac{\partial^2 v}{\partial x^2} + (k - 1) \frac{\partial v}{\partial x} - kv \quad (8)$$

where $k = 2r/\sigma^2$.

Step 2: Method of Undetermined Coefficients (Week 2). Assume a solution of the form:

$$v(x, \tau) = e^{\alpha x + \beta \tau} u(x, \tau) \quad (9)$$

Substitute into (8), collect terms, and choose α and β to make the $\frac{\partial u}{\partial x}$ and u terms vanish:

$$\alpha = -\frac{k-1}{2}, \quad \beta = -\frac{(k+1)^2}{4} \quad (10)$$

The result is the **Heat Equation**:

$$\boxed{\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}} \quad (11)$$

with initial condition:

$$u(x, 0) = \max \left(e^{\frac{k+1}{2}x} - e^{\frac{k-1}{2}x}, 0 \right) \quad (12)$$

Syllabus Connections:

- **Week 1 — Wronskian:** The two fundamental solutions $e^{\sqrt{s}x}$ and $e^{-\sqrt{s}x}$ are linearly independent, verified by a non-zero Wronskian.
- **Week 2 — Dimensionality:** The solution space of the heat equation with one initial condition is one-dimensional.

2.4 Solution via Laplace Transform (Week 3)

Apply the Laplace transform in τ : $\hat{u}(x, s) = \int_0^\infty e^{-s\tau} u(x, \tau) d\tau$. The heat equation becomes:

$$\frac{d^2 \hat{u}}{dx^2} - s\hat{u}(x, s) = -u(x, 0) \quad (13)$$

This is a second-order linear ODE with constant coefficients. The homogeneous solution is:

$$\hat{u}_h(x, s) = A(s)e^{-\sqrt{s}x} + B(s)e^{\sqrt{s}x} \quad (14)$$

For $x > 0$, we find a particular solution using undetermined coefficients. Applying the boundary conditions (boundedness as $x \rightarrow \pm\infty$, continuity at $x = 0$) and taking the inverse Laplace transform yields the famous **Black–Scholes Formula**:

$$\boxed{V(S, t) = SN(d_1) - Ke^{-r(T-t)}N(d_2)} \quad (15)$$

where:

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}} \quad (16)$$

$$d_2 = d_1 - \sigma\sqrt{T - t} \quad (17)$$

and $N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz$ is the standard normal CDF.

3 Worked Example: Calculating an Option Price

Let's use the formula with real numbers.

Given Data:

- Current stock price: $S = 100$
- Strike price: $K = 100$
- Time to expiry: $T - t = 1$ year
- Risk-free rate: $r = 5\% = 0.05$
- Volatility: $\sigma = 20\% = 0.20$

Step 1: Calculate d_1 .

$$d_1 = \frac{\ln(100/100) + (0.05 + 0.20^2/2) \times 1}{0.20 \times \sqrt{1}} \quad (18)$$

$$= \frac{0 + (0.05 + 0.02) \times 1}{0.20} \quad (19)$$

$$= \frac{0.07}{0.20} \quad (20)$$

$$= 0.35 \quad (21)$$

Step 2: Calculate d_2 .

$$d_2 = d_1 - \sigma\sqrt{T-t} \quad (22)$$

$$= 0.35 - 0.20 \times 1 \quad (23)$$

$$= 0.15 \quad (24)$$

Step 3: Find $N(d_1)$ and $N(d_2)$.

Using a standard normal table (or calculator):

$$N(d_1) = N(0.35) \approx 0.6368 \quad (25)$$

$$N(d_2) = N(0.15) \approx 0.5596 \quad (26)$$

Step 4: Plug into the Black–Scholes formula.

$$V = S \times N(d_1) - K \times e^{-r(T-t)} \times N(d_2) \quad (27)$$

$$= 100 \times 0.6368 - 100 \times e^{-0.05} \times 0.5596 \quad (28)$$

$$= 63.68 - 100 \times 0.9512 \times 0.5596 \quad (29)$$

$$= 63.68 - 53.23 \quad (30)$$

$$= 10.45 \quad (31)$$

Answer: The fair price of this European call option is approximately **Rs. 10.45**.

Interpretation: If you buy this option for Rs. 10.45:

- If the stock ends above Rs. 110.45 (strike + premium), you make a profit.
- If the stock ends between Rs. 100 and Rs. 110.45, you recover part of your premium.
- If the stock ends below Rs. 100, you lose the entire Rs. 10.45 premium.

4 The Problem: Volatility Smile

4.1 Implied Volatility

The Black–Scholes formula takes σ as input. We can also go backwards: plug in the market price V_{market} and solve for the **implied volatility** σ_{imp} .

If the model were perfect, σ_{imp} should be the **same** for all strike prices K .

4.2 What the Market Actually Shows

When we plot σ_{imp} against K for real market data, we get:

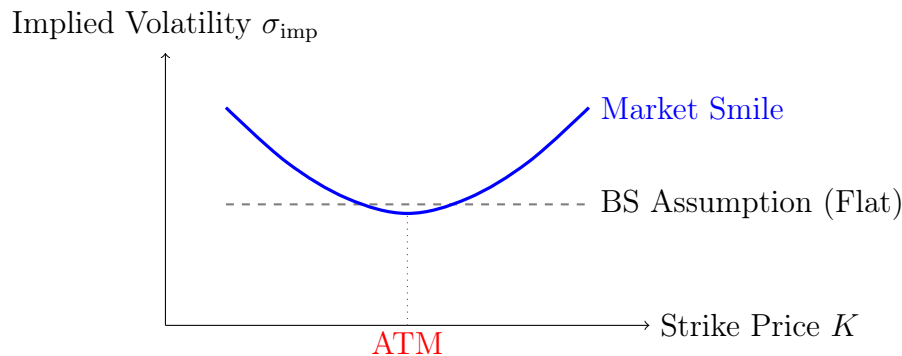


Figure 1: The Volatility Smile: Market data vs. the Black–Scholes flat-line assumption.

This smile shape tells us: The market does **not** believe volatility is constant. Out-of-the-money and in-the-money options require higher volatility inputs, meaning the market expects more extreme price moves than the lognormal distribution predicts.

4.3 Why This Breaks the Analytical Approach

If volatility is not constant but depends on S and t (or is itself random):

- The reduction to the heat equation **no longer works** — coefficients become state-dependent.
- The Laplace transform method **fails** — the resulting ODE would have non-constant coefficients.
- We need a different mathematical tool.

5 Monte Carlo Simulation: A Different Approach

5.1 The Key Idea (Feynman–Kac Theorem)

There is a powerful connection between PDEs and expectations. The Feynman–Kac theorem states that instead of solving the Black–Scholes PDE, we can compute:

$$\boxed{V(S, t) = e^{-r(T-t)} \times \mathbb{E} [\max(S_T - K, 0)]} \quad (32)$$

In simple words: **The option price is the average discounted payoff over all possible future stock prices.**

5.2 How Monte Carlo Works

1. **Start** with today's stock price $S_0 = 100$.

2. **Simulate one path:** Divide 1 year into small steps (e.g., 252 trading days). At each step, the price changes randomly:

$$S_{\text{new}} = S_{\text{old}} \times e^{(r-\sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}Z} \quad (33)$$

where Z is a random number from the standard normal distribution $N(0, 1)$.

3. **At expiry**, calculate the payoff: $\max(S_T - 100, 0)$.
4. **Repeat** steps 2–3 many times (e.g., 100,000 paths).
5. **Average** all payoffs and discount back to today:

$$\text{Price} = e^{-rT} \times \frac{\text{Sum of all payoffs}}{\text{Number of paths}} \quad (34)$$

5.3 Example with Our Numbers

Using the same data as Section 3 ($S = 100$, $K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.20$):

- **Path 1:** Random numbers lead to final price $S_T = 108$. Payoff = $\max(108 - 100, 0) = 8$.
- **Path 2:** Random numbers lead to $S_T = 95$. Payoff = $\max(95 - 100, 0) = 0$.
- **Path 3:** Random numbers lead to $S_T = 115$. Payoff = $\max(115 - 100, 0) = 15$.
- ... and so on for 100,000 paths.

After 100,000 simulations, the average discounted payoff converges to approximately **Rs. 10.45** — matching the Black–Scholes formula result.

5.4 Why Monte Carlo is Powerful

Analytical PDE Approach	Monte Carlo Approach
Needs transformation to heat equation	Needs only the SDE
Uses Laplace transforms, undetermined coefficients	Uses random numbers and averaging
One exact number	Estimate with confidence interval
Fails if σ is not constant	Works with any volatility model

Table 1: Comparison of the two approaches.

The key advantage: if volatility is not constant — for example, if it follows the Heston model where σ itself is random — Monte Carlo handles it naturally by adding one more equation. The analytical PDE approach would require solving a two-dimensional PDE with no simple closed form.

6 Conclusion

This project demonstrated two ways to model option pricing:

1. **The Black–Scholes PDE approach** uses techniques from our full differential equations syllabus:
 - **Week 1:** Picard’s theorem (existence and uniqueness), Wronskian (linear independence).
 - **Week 2:** Constant-coefficient ODEs, method of undetermined coefficients, dimensionality of solution space.
 - **Week 3:** Boundary value problems, Laplace transform method.

The result is an elegant closed-form formula. We demonstrated it with a worked example: for $S = 100$, $K = 100$, $T = 1$, $r = 5\%$, $\sigma = 20\%$, the call option price is Rs. 10.45.

2. **The volatility smile** reveals that real markets reject the constant volatility assumption. When σ is not constant, the analytical approach breaks down.
3. **Monte Carlo simulation** offers a robust alternative. By simulating many random price paths and averaging their payoffs, we can price options under any volatility model. The method is simple, flexible, and handles real-world complexities that defeat analytical PDE solutions.

Final Message: Mathematical modeling is not just about finding beautiful formulas. It is about knowing when those formulas work, when they fail, and having the courage to switch to a different tool when reality demands it.

References

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